

OmniGesture v2.0

AI-Powered Hands-Free PC Controller

Built for Snapdragon-Powered HP PCs

Snapdragon AI Lab Build & Present Challenge | September 2026

The Problem

Standard PC navigation is not accessible to everyone

- >> **1.3B+ people worldwide**
live with disabilities affecting motor function
- >> **Assistive hardware costs \$1,500-\$10,000+**
making it inaccessible in developing economies
- >> **Cloud AI tracking has 200-500ms latency**
making real-time cursor control impossible
- >> **Streaming video to cloud servers**
creates massive privacy violations
- >> **Heavy GPU models drain batteries**
and cause thermal throttling on laptops

The Solution: OmniGesture v2.0

Turn your webcam into a zero-latency gesture controller

Point (Index Up)

Move Cursor

Pinch (Thumb+Index)

Left Click

Peace (Index+Middle)

Double Click

Fist (All Closed)

Right Click

Open Palm (All Up)

Scroll Up/Down

Thumbs Up

Toggle Drag Lock

Plus: Dark-themed Control Panel | Live FPS Counter | Sensitivity Slider | Activity Log

Why Snapdragon?

Edge AI is not optional for real-time gesture control - it is mandatory

	Cloud AI	Snapdragon Edge AI
Latency	200-500ms (unusable)	<15ms (fluid)
Privacy	Video sent to servers	100% on-device
Internet	Required always	Fully offline
Battery	Heavy drain	NPU-optimized
Cost	Per-API-call charges	Free forever

Technical Architecture

HP Webcam --> OpenCV 4.10 --> MediaPipe HandLandmarker --> Gesture Engine --> PyAutoGUI --> OS Mouse Events

MediaPipe Tasks API

Open-source edge AI model detecting 21 hand landmarks at 30+ FPS

Gesture Recognition

Finger state vectors (tip vs PIP elevation) + Euclidean pinch distance

EMA Cursor Smoothing

Exponential Moving Average filter eliminates tremors and micro-jitter

Multi-threaded Design

Vision engine runs in background thread; Tkinter UI on main thread

Temporal Cooldowns

State machine prevents accidental double-triggers during transitions

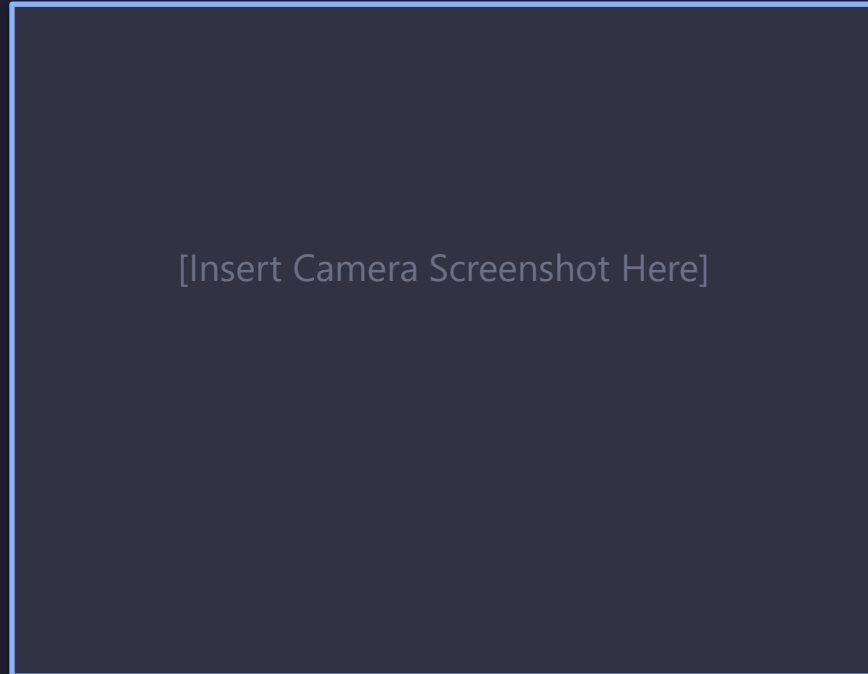
Adjustable Sensitivity

User-configurable smoothing factor via slider (1-10 range)

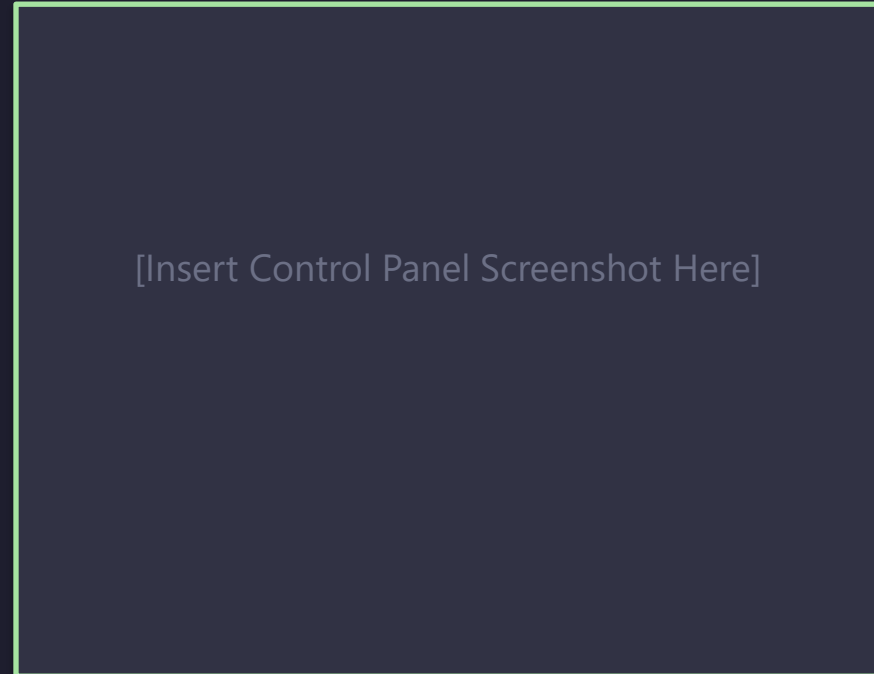
Tech Stack: Python 3.10+ | MediaPipe (Open Source) | OpenCV 4.10 | PyAutoGUI | Tkinter

Live Demo & Screenshots

Camera View (Hand Tracking HUD)



Tkinter Control Panel



Both windows run simultaneously - Camera HUD shows landmarks, Control Panel shows status

Future Roadmap & Vision

Phase 1: Qualcomm AI Hub NPU Deployment

Compile and INT8-quantize the hand landmark model for the Snapdragon Hexagon NPU via Qualcomm AI Hub. Target: sub-watt inference with near-zero CPU overhead.

Phase 2: Custom Gesture Macro Builder

Allow users to record custom hand poses and map them to any keyboard shortcut or system action (e.g., Peace Sign = Copy, Swipe Left = Switch Window).

Phase 3: Multimodal Head & Gaze Tracking

Add head-pose estimation and iris gaze tracking for users with severe mobility impairments who cannot use their hands at all (e.g., quadriplegia).

OmniGesture makes every Snapdragon HP PC universally accessible - no extra hardware, no cloud, no compromise.

Thank You! | GitHub: github.com/sumitbodal/OmniGesture